

Verilog

1) Complete the table with equivalents in Binary, Octal and hexa for 143.75 (decimal), 11011.101 (Binary), 453 (Octal), 1F.C (Hex)

Decimal	Binary	Octal	Hexa
143.75	10001111.11	217.6	8F.C
27.625	11011.101	33.5	1B.A
299	100101011	453	12B
31.75	11111.11	37.6	1F.C

now $143.75_{(10)} \rightarrow (?)_{(2)}$

2	143	
2	71	-1
2	35	-1
2	17	-1
2	8	-1
2	4	-0
2	2	-0
	1	-0

$$0.75 \times 2 = 1.5 - 1$$

$$0.5 \times 2 = 1 - 1 \downarrow$$

$$143.75_{10} = \underline{10001111.11}_2$$

now $143_{(10)} \rightarrow (?)_8$

$$\begin{array}{r} 8 \overline{) 143} \\ 8 \overline{) 17} - 7 \\ \hline 2 - 1 \end{array}$$

$$0.75 \times 8 = 6.0 = 6$$

$$143_{(10)} = 217.6_{(8)}$$

now $143_{(10)} \rightarrow (?)_{16}$

$$\begin{array}{r} 16 \overline{) 143} \\ \hline 8 - 15 \end{array}$$

$$0.75 \times 16 = 12.0$$

$$\therefore 143_{(10)} = 8F_{(16)}$$

now $11011.101_{(2)} \rightarrow (?)_{10}$

$$1 \times 2^4 + 1 \times 2^3 + 0 + 1 \times 2 + 1 \times 2^0 + 1 \times 2^{-1} + 0 + 1 \times 2^{-3} \\ = 27.625_{(10)}$$

now $11011.101 \rightarrow (?)_8$

$$\underbrace{011}_3 \underbrace{011}_3 \underbrace{.101}_{.5} = 33.5_{(8)}$$

now $\underbrace{00011}_5 \underbrace{011}_{.75} \underbrace{.1010}_{.625} = 18.75_{(16)}$

now $453_{(8)} \rightarrow (?)_{10}$

$$4 \times 8^2 + 5 \times 8^1 + 3 \times 8^0 = 299_{(10)}$$

now $453_{(8)} \rightarrow (?)_2$

$$\underbrace{100}_4 \underbrace{0101}_{13} \rightarrow 100101011_{(2)}$$

now $453_{(8)} \rightarrow (?)_{16}$

1	1
1	25
1	41
1	8
0	+
0	15
0	1

$$\underline{0001} \underline{0010} \underline{1011} = 12B$$

$$\text{now } 1F_{(10)} \rightarrow (?)_{(10)}$$

$$1 \times 16^1 + 15 \times 16^0 + 12 \times 16^{-1} = 31.75_{(10)}$$

$$\text{now } 1F_{(16)} \rightarrow (?)_2$$

$$= 00011111.1100_{(2)}$$

$$\text{now } 1F_{(10)} \rightarrow (?)_8$$

$$\underline{011111} \underline{110} = 37.6_{(8)}$$

2) Find equivalents of the following

$$1) 324_{(5)} \rightarrow (?)_{(10)}$$

$$= 3 \times 5^2 + 2 \times 5^1 + 4 = 89_{(10)}$$

$$2) 32.23_{(4)} \rightarrow (?)_{(10)}$$

$$= 3 \times 4^1 + 2 + 2 \times 4^{-1} + 3 \times 4^{-2} = 14.6875_{(10)}$$

$$3) 231_{(10)} = ?_{(7)}$$

$$\begin{array}{r} 7 \overline{) 231} \\ 7 \overline{) 23} - 0 \\ \hline 4 - 5 \end{array} \quad \begin{array}{l} = 450_{(7)} \\ \rightarrow \end{array}$$

$$4) 189_{(10)} = (?)_{(6)}$$

$$\begin{array}{r} 6 \overline{) 189} \\ 6 \overline{) 31} - 3 \\ \hline 5 - 1 \end{array} \quad \begin{array}{l} = 513_{(6)} \\ \rightarrow \end{array}$$

③ Find the radii r of following - 110101001000

1) $(312/20)r = (13-1)r$

$$= \frac{312r}{20r} = 13-1r$$

$$= \frac{3 \times r^2 + 1 \times r + 2}{2 \times r + 0} = 1 + r + 3 + 1 \times r^{-1}$$

$$= \frac{3r^2 + r + 2}{2r} = r + 3 + \frac{1}{r}$$

$$= 3r^2 + r + 2 = 2r^2 + 6r + 2$$

$$= r^2 - 5r = 0$$

$$= r^2 - 5r$$

$$= r = 5$$

2) $264r + 136r = 433r$

$$= 2 \times r^2 + 6 \times r + 4 + 1 \times r^2 + 3 \times r + 6 = 4 \times r^2 + 3 \times r + 3$$

$$= 3r^2 + 9r + 10 = 4r^2 + 3r + 3$$

$$= -r^2 + 6r + 7 = 0$$

$$r = -1 \quad r = 7$$

r cannot be negative

so $r = 7$

$$3) 221505_{(8)} = 12345_{(8)}$$

$$= 2 \times 8^5 + 2 \times 8^4 + 1 \times 8^3 + 5 \times 8^2 + 0 + 5 \times 1 = 1 \times 8^4 + 2 \times 8^3 + 3 \times 8^2 + 4 \times 8 + 5$$

$$= 8 - 16$$

4) Addition

$$1) 1101_{(2)} + 1101111_{(2)}$$

$$\begin{array}{r} 1101 \\ 1101111 \\ \hline 11101100 \end{array} = 11101100_{(2)}$$

$$2) 77652_{(8)} + 76543_{(8)}$$

$$\begin{array}{r} 77652 \\ 76543 \\ \hline 176415 \end{array} = 176415_{(8)}$$

$$3) 578A_{(16)} + ABDE_{(16)}$$

$$\begin{array}{r} 578A \\ ABDE \\ \hline 10368 \end{array}$$

$$A + E = 24_{(10)} = 16 \overline{)24} = (18)_{(16)} \quad 1-8$$

$$9 + D = 22_{(10)} = 16 \overline{)22} = 16_{(16)} \quad 1-6$$

$$8 + B = 19_{(10)} = 16 \overline{)19} = 13_{(16)} \quad 1-3$$

$$6 + A = 16_{(10)} = 16 \overline{)16} = 10_{(16)} \quad 1-0$$

$$4) 234(10) + 315(10)$$

$$\begin{array}{r} 234 \\ 315 \\ \hline 553 \end{array}$$

$$= 553(10)$$

$$5 + 4 = 9(10) = 6 \begin{array}{l} 9 \\ 1 \end{array} - 3 = 13$$

$$3 + 2 = 5(10) = 6 \begin{array}{l} 5 \\ 0 \end{array} - 5 = 05$$

5) Binary - Gray code (conversion XOR operation / bit-wise XOR)

$$1) 10101(2) = 10101(2)$$

$$\rightarrow 1111(2)$$

$$2) 01110(2) = 01110(2)$$

$$\rightarrow 01001(2)$$

6) Gray code - Binary conversion

$$1) 01101(\text{gray})$$

$$\rightarrow 01101(\text{gray}) = [01001(2)]$$

$$\rightarrow 01001(2)$$

$$2) 10101(\text{gray})$$

$$\rightarrow 10101(\text{gray}) = [11001(2)]$$

$$\begin{array}{r} \text{XOR} \quad \text{XOR} \quad \text{XOR} \quad \text{XOR} \\ 1 \quad 0 \quad 1 \quad 0 \quad 1 \\ \hline 1 \quad 1 \quad 0 \quad 0 \quad 1 \end{array} (2)$$